Routely

Online bus ticket booking platform - project report

1. Overview

Routely lets passengers search bus routes, compare departures, pick exact seats and receive a ticket in under a minute. The prototype is a single-page web app (HTML, CSS, JavaScript) with a route map, live seat map and booking confirmation. It is designed to grow into a full startup product with operator dashboards and online payments.

2. Problem and opportunity

- Travellers still rely on agents, phone calls and crowded counters.
- Small operators have no simple way to sell seats online.
- Seat double-booking and lost tickets cause refunds and complaints.

3. Core features (prototype)

- Route search by origin, destination and date, with multi-stop routes found automatically.
- Interactive route map highlighting the chosen path.
- Three bus types per route (Seater, Sleeper, AC Sleeper) with distance-based fares.
- Seat map (2+2 layout), up to 6 seats per booking, booked seats become unavailable.
- Passenger details with validation, printable ticket with booking ID.

4. Booking flow

Search route → Pick a bus → Choose seats → Passenger details → Ticket issued (seats locked).

5. Route network (sample data)

From	To	Distance	Duration	Base fare (Rs)
Delhi	Jaipur	280 km	5.5 h	450
Delhi	Agra	230 km	4 h	380
Delhi	Lucknow	550 km	8 h	700
Agra	Lucknow	330 km	5 h	420
Jaipur	Ahmedabad	660 km	10 h	850
Agra	Bhopal	470 km	8 h	600
Bhopal	Nagpur	350 km	6 h	480
Ahmedabad	Mumbai	530 km	8 h	720

6. Production architecture

• Frontend: this web app, later a React or mobile app.
• Backend API: Node.js or Python, REST endpoints for search, seat hold, booking.
• Database: PostgreSQL. Seat locking with a short hold (about 10 minutes) to avoid double booking.
• Payments: UPI and card gateway, with SMS/email ticket delivery.
• Operator portal: add routes, schedules, prices and view sales.

7. Database design

Table	Key fields
operators	id, name, phone, commission
buses	id, operator_id, type, seat_count
routes	id, origin, destination, distance_km
trips	id, route_id, bus_id, depart_at, fare
bookings	id, trip_id, passenger_name, phone, status, total
booking_seats	booking_id, trip_id, seat_no (unique per trip)

8. Business model

• Commission per ticket from operators (8-12%).
• Convenience fee for passengers.
• Featured listings and promotions for operators.

9. Roadmap

Phase	Scope
MVP (month 1-2)	Working prototype, 2-3 pilot operators on one corridor
Launch (month 3-5)	Backend, payments, SMS tickets, operator portal
Growth (month 6-12)	Mobile apps, more cities, cancellations and refunds, ratings

10. Risks

• Operator onboarding is the hardest part: start with one busy corridor.
• Payment failures and seat holds need careful handling.
• Competition from large aggregators: win on simple UX and fair commissions.